FIN — Release 10.32

Research Programs P365–P378

Exact certificate diversity, adaptive discrimination,
operational resource semantics, and the universality boundary

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31 July 2026

Publication — Preprint; Version 1.0.0

Tightened oscillatory resource.

$$0.70727845855518420 \leq \mathcal{N}_{\mathrm{osc}} \leq 0.70735346839982 60.$$

Ideal adaptive discrimination, first perfect branch.

$$\frac{1}{2}\left\|\mathcal{U}_{\mathrm{strict}}^{\otimes n} - \mathcal{U}_{\mathrm{legacy}}^{\otimes n}\right\|_{\diamond} = \sin\left(\frac{nt\Delta}{2}\right), \qquad 0 \leq t \leq \frac{\pi}{n\Delta}.$$

The signed quantity is a classical representation cost. The channel theorem is restricted to the declared commuting, ideal-unitary comparison and its first perfect time. No selector, dimensional standard, laboratory realization, physical role transfer, Standard Model, gravity, or Theory-of-Everything closure is claimed.

Repository. `https://github.com/hyconiek/Fractal-Nadsoliton-Theory`

Publication statement

This monograph reports every executed Program P365–P378 and recommends Programs P379–P393. Exact proof, high-precision computation, synthetic apparatus modelling, conditional operational semantics, and absent external evidence are kept distinct. The legacy/strict split and all current strict-core guardrails remain binding.

Contents

Confidence convention

Label	Meaning
[Proven]	Exact mathematical proof, exact rational computation, exhaustive finite classification, or compiled structural formal theorem
[Strong evidence]	Reproducible high-precision computation with explicit residual and robustness boundary
[Moderate evidence]	Model-dependent synthetic or local result with stated assumptions
[Conditional]	Result requiring named encodings, axioms, standards, or operational identifications
[Refuted]	The declared proposition fails in its stated class
[Blocked by external evidence]	Independent data, hardware, calibration, standards, custody, or unblinding is required

Chapter 1

Scientific scope and guardrails

1.1 Kernel split

The legacy intermediate kernel remains

$$K_{\text{legacy,ont}}(d) = 4\ln 2\,\frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d},$$

whereas the strict working kernel is

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

No identity between these functions is assumed. The scale $\gamma = 0.13044237640961015$ used in P371 is the frozen operational comparison scale from P328/P343, not a bridge-completion or physical role-transfer theorem.

1.2 Ontology

The repository order remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton is the primordial information in a solitonic state, not an object above a second informational substrate. None of the proofs below uses this ontology as a mathematical premise.

1.3 Preserved open boundaries

This report does not close:

- QW-2191 or the non-premise selector;
- the legacy-to-strict completion;
- legacy physical-role transfer;
- an internal length, time, action, energy, or mass standard;
- a laboratory realization functor;
- a unit-bearing nonproxy total action;
- Standard Model or gravity generation;

- Theory-of-Everything closure.

Chapter 2

Executive results

Program	Result	Status
P365	Independent exact checker reproduces P351/ P352 identities	[Proven] / proof-assistant arithmetic still blocked
P366	Oscillatory enclosure tightened to $[0.7072784586, 0.7073534684]$	[Proven]
P367	No independent QW hold-out admitted	[Blocked]
P368	Maximal k-distance quotient category; exhaustive small-graph classification	[Proven]
P369	Complex transfer locally full-rank; intensity calibration ill-conditioned	[Moderate evidence]
P370	No calibrated photonic pilot admitted	[Blocked]
P371	Full ideal adaptive optimum through the first perfect time	[Proven]
P372	No traceable clock anchor admitted	[Blocked]
P373	Five operational data-processing monotones	[Proven] / [Conditional] FIN encoding
P374	Noncanonical outer analytic extension family	[Proven]
P375	No conditional EW blind-test bundle admitted	[Blocked]
P376	No dimensional standards bundle admitted	[Blocked]
P377	FORC resource syntax free; unique operational universality false	[Proven] / [Refuted]
P378	No reservoir tomography bundle admitted	[Blocked]

Chapter 3

P365 — independent checking is not self-replay

3.1 Checker separation

The P365 checker:

- imports no FIN research module;
- reimplements rational intervals;
- reimplements matrix inversion over $\mathbb{Q}$;
- reimplements Krawczyk inclusion;
- reimplements fifth-root brackets;
- reimplements Taylor/Lagrange cosine intervals;
- reimplements Bernstein conversion and subdivision;
- recomputes the exact Vandermonde inverse;
- compares byte-level rational identities with archived certificates.

3.2 Results

The checker reproduces:

Component	**Checks**
P351 Krawczyk interval endpoints	12
P352 safe dual coefficients	12
P352 weight interval endpoints	24
P352 Bernstein feasibility	passed
P352 twelve weight signs	passed

All checks pass.

3.3 Confidence boundary

[Proven] A second implementation using exact arbitrary-precision arithmetic accepts the certificates.

[**Blocked**] The large arithmetic is not yet reflected through a Lean or Coq kernel. The compiled Lean file proves structural implications only.

This distinction matters. Independent implementation reduces the risk of a shared code-path error; proof-assistant reflection would additionally reduce the trusted computing base.

Chapter 4

P366 — a tighter seven-atom oscillatory certificate

4.1 Exact dual refinement

P352 used 4096 rational Bernstein subintervals. P366 uses a depth-fourteen dyadic de Casteljau tree:

$$2^{14} = 16\,384$$

subintervals.

For the rational degree-eleven dual candidate p_{raw}, exact Bernstein extrema L, U define

$$p_{\mathrm{safe}}(x) = \frac{p_{\mathrm{raw}}(x) - U}{U - L}.$$

The same dyadic tree proves

$$-1 \le p_{\mathrm{safe}}(x) \le 0 \qquad (0 \le x \le 1).$$

Exact oscillatory moment intervals then give

$$\mathcal{N}_{\mathrm{osc}} \ge 0.70727845855518420.$$

4.2 Seven-atom primal

The first node is fixed exactly at

$$r_1 = \frac{19826532267005523}{10^{18}}.$$

The remaining unknowns are:

- five interior nodes;
- seven weights.

All twelve moment equations are included. The exact rational Krawczyk image is strictly contained in a radius-10^{-20} box for all twelve variables. The certified sign pattern is

$$(-,+,+,+,+,-,+).$$

The corresponding feasible negative mass satisfies

$$\mathcal{N}_{\mathrm{osc}} \le 0.7073534683998260.$$

4.3 Result

$$\boxed{0.7072784585518420 \le \mathcal{N}_{\mathrm{osc}} \le 0.7073534683998260}.$$

Compared with P352:

- the lower endpoint improves by 2.3323561×10^{-7};
- the upper endpoint improves by 6.5064864×10^{-6};
- the width decreases by 8.244%.

4.4 Interpretation boundary

This is the minimum negative part of a signed scalar representing measure in the declared truncated Hausdorff problem. It is not automatically:

- quantum quasiprobability;
- negative information;
- dissipated entropy;
- negative energy;
- an adaptive feedback strength;
- a physical coupling.

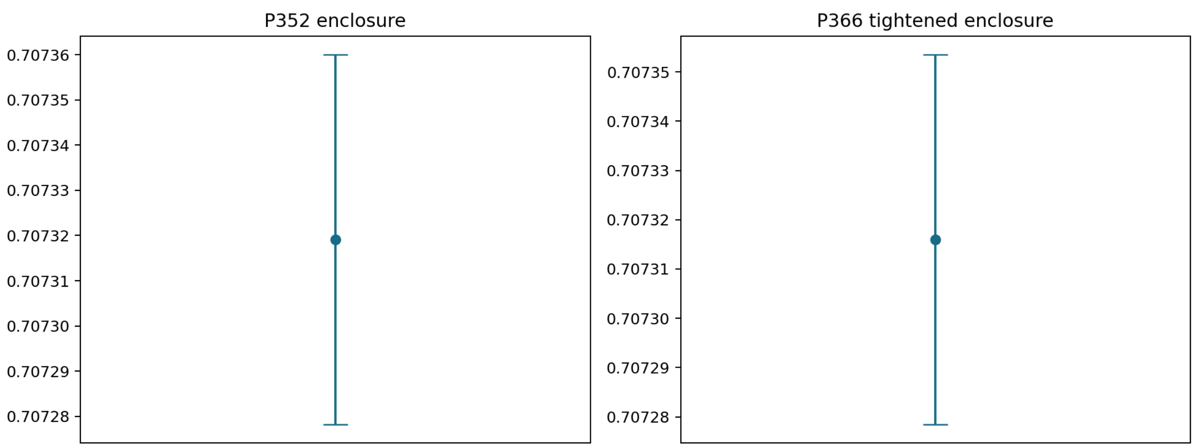

Figure 4.1: Independent exact certificate checking and the tightened oscillatory signed-moment enclosure.

Chapter 5

P367 — external hold-out gate

No manifest satisfies the independent provider/registrar/analyst, freeze-hash, and one-shot unblinding contract.

[**Blocked by external evidence**]

The continuous QW refreeze remains in-sample. Its numerical stability cannot be promoted to an empirical prediction.

Chapter 6

P368 — the maximal radial naturality category

6.1 Exact categorical definition

Let G, H be finite metric carriers and k a radial law. A map $f : G \to H$ preserves the radial kernel exactly when

$$k(d_H(f(x), f(y))) = k(d_G(x, y))$$

for all x, y.

Define the distance equivalence relation

$$d \sim_k e \iff k(d) = k(e).$$

Then f is kernel-preserving precisely when it preserves the $\sim_k$-class of every pairwise distance.

These maps contain identities and are closed under composition. They form the maximal category on which the radial kernel is strictly natural.

6.2 Injective-distance corollary

If k is injective on the attainable distance set, then

$$k(d_H(fx, fy)) = k(d_G(x, y)) \iff d_H(fx, fy) = d_G(x, y).$$

Thus the maximal naturality category reduces to isometric embeddings.

6.3 Exact small-distance certificate

Exact rational intervals for

$$k_{\text{strict}}(1), \dots, k_{\text{strict}}(4)$$

are pairwise disjoint. Consequently the strict kernel is injective on the distance sets of connected graphs with at most five vertices.

6.4 Exhaustive atlas

The finite audit covers all thirty connected unlabeled graphs with two to five vertices.

Quantity	**Count**
Graph pairs admitting at least one homomorphism	607
Graph homomorphisms	77,130
Isometric embeddings	1,664
Strict-kernel-preserving maps	1,664
Classification mismatches	0

6.5 Boundary

No global theorem says the oscillatory law is injective for every integer distance. Larger carriers must use the k-distance quotient unless further distance intervals are certified.

This result classifies carrier transport. It does not construct a legacy-to-strict completion law.

Chapter 7

P369–P370 — component losses and the photonic boundary

7.1 Model

The ideal twelve-mode unitary has 66 Givens components. P369 assigns:

- one logarithmic insertion-loss parameter per component;
- one phase-error parameter per component.

The unknown parameter count is therefore

$$2 \times 66 = 132.$$

7.2 Local identifiability

At the ideal point, finite-difference Jacobians give:

Observation	**Effective rank**	**Condition number**
Complex transfer matrix	132	1.94×10^2
Intensities $\lvert T_{ij}\rvert^2$	123	5.66×10^7
Conditional intensities	118	8.32×10^7

The threshold is 10^{-8} relative to the largest singular value.

[Moderate evidence]

Complex transfer tomography is locally informative for all declared parameters. Intensity-only observations possess practically unresolved directions and severe conditioning.

7.3 Correlated-error stress test

The synthetic model varies:

- loss from 0.001 to 0.01 dB per component;
- phase noise from 10^{-4} to 10^{-3} radians;
- phase AR(1) correlation 0 or 0.8;
- 64 runs per setting.

The worst tested setting has

$$\mathrm{TV}_{95} = 0.0051152391,$$

and fifth-percentile minimum click probability

$$0.9606933924.$$

7.4 What this changes experimentally

A credible photonic pilot must freeze a complex transfer calibration, not only vertex intensity images. It must also report:

- phase-reference stability;
- source coherence and multiphoton contamination;
- detector efficiency and crosstalk;
- raw event timestamps;
- calibration uncertainty;
- independent custody.

P370 finds no admitted physical bundle.

[Blocked by external evidence]

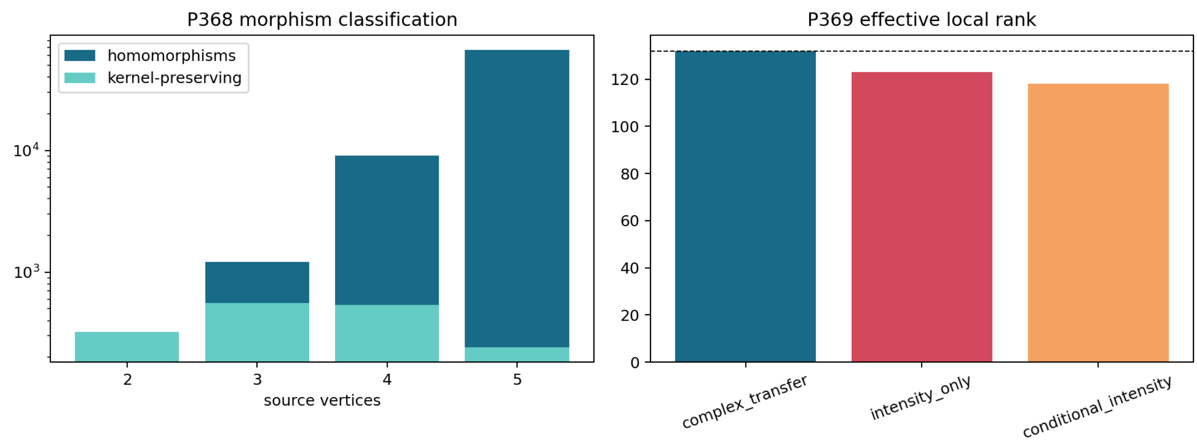

Figure 7.1: The maximal radial-kernel naturality category and component-level photonic calibration identifiability.

Chapter 8

P371 — exact adaptive-angle theorem

8.1 Relative generator

At the frozen comparison scale γ, define

$$D = \gamma A_{\text{legacy}} - A_{\text{strict}}.$$

The two circulant generators commute to numerical norm

$$4.22 \times 10^{-15}.$$

Their relative unitary is therefore

$$U_{\text{strict}}(t)^{\dagger} U_{\text{legacy}}(t) = e^{-itD}.$$

Let

$$\Delta = \lambda_{\max}(D) - \lambda_{\min}(D) = 4.365523996749653.$$

8.2 Adaptive upper bound

Consider arbitrary coherent memory, ancillas, and common adaptive controls. Before an oracle call, let the two hypothesis branches have Fubini–Study angle Θ. Applying different unitary hypotheses can add at most

$$\frac{t\Delta}{2}$$

to the angle. Common controls preserve the angle. By induction and the triangle inequality,

$$\Theta_n \leq \frac{nt\Delta}{2}.$$

For pure output branches, the half trace distance is bounded by

$$\sin \Theta_n.$$

This gives an unrestricted adaptive upper bound.

8.3 Parallel lower bound

Let $|\lambda_{\min}\rangle$ and $|\lambda_{\max}\rangle$ be extremal eigenvectors of D. A parallel n-use superposition of the two tensor extremal modes has overlap

$$\cos\left(\frac{nt\Delta}{2}\right).$$

It therefore attains half trace distance

$$\sin\left(\frac{nt\Delta}{2}\right).$$

The adaptive upper bound and parallel lower bound coincide.

8.4 Theorem

For

$$0 \le t \le \frac{\pi}{n\Delta},$$

the unrestricted adaptive optimum is

$$\boxed{D_n(t) = \sin\left(\frac{nt\Delta}{2}\right)}.$$

At

$$t_n^* = \frac{\pi}{n\Delta},$$

the two channels are perfectly distinguishable.

Uses n	t_n^*
1	0.7196370140053893
2	0.3598185070026946
3	0.2398790046684631
4	0.1799092535013473

The offsets between these values and the earlier P343 sampled grid are all below 3.63×10^{-4}.

8.5 Scope

The theorem solves the ideal commuting unitary pair through the first perfect-discrimination time. It assumes:

- identical calibrated time per use;
- the frozen comparison scale;
- lossless unitary calls;
- arbitrary coherent memory and preparation;
- unrestricted final measurement.

It does not solve:

- lossy or dephasing channels;
- uncertain or nonlinear clocks;
- limited ancilla dimensions;
- imperfect instruments;
- the global post-threshold revival structure.

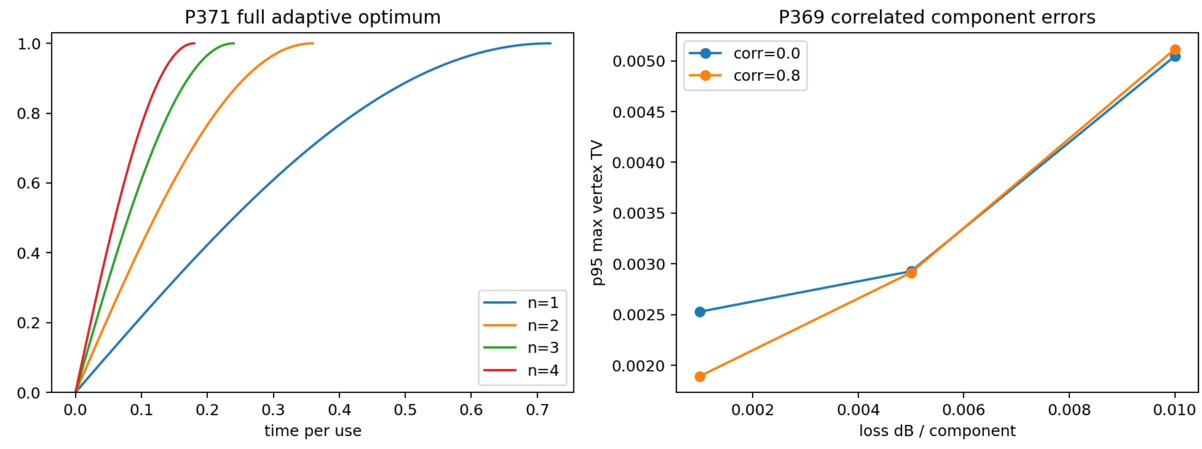

Figure 8.1: Synthetic component-loss sensitivity and the exact ideal adaptive discrimination law through the first perfect time.

Chapter 9

P372 — the clock remains external

P358 proved an equispaced minimax design after a dimensional anchor is supplied. P372 searches for a manifest containing:

- traceable clock-standard hash;
- calibration-record hash;
- provider, registrar, and analyst;
- freeze timestamp;
- unblinding authorization.

No admissible package is present.

[**Blocked by external evidence**]

P371's analytic time parameter is therefore dimensionless repository time, not a calibrated laboratory second.

Chapter 10

P373 — operational semantics for five resources

10.1 Signed-path resource

For a signed measure of fixed total mass,

$$\mathcal{N}(\mu) = \mu^{-}(X)$$

contracts under positive mass-preserving Markov pushforwards.

10.2 Phase resource

For a density operator,

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|$$

contracts under the declared dephasing channel.

This is an operational coherence resource. It is not identical to the number-theoretic label "non-torsion phase" without an explicit encoding.

10.3 Orientation resource

For a distribution p on $\mathbb{Z}_{12}$ and reflection R, define

$$A_R(p) = \frac{1}{2}\|p - Rp\|_1.$$

Reflection-covariant Markov convolution commutes with R and contracts total variation, so

$$A_R(Tp) \leq A_R(p).$$

This measures an already prepared orientation asymmetry. It does not source the missing strict selector.

10.4 Scale resource

For a parametric distribution p_θ, classical Fisher information

$$I(\theta) = \sum_x \frac{(\partial_\theta p_\theta(x))^2}{p_\theta(x)}$$

contracts under a parameter-independent Markov coarse graining.

This quantifies information about an existing scale parameter. It does not create a dimensional standard.

10.5 Cross-law resource

For a joint distribution,

$$I(X : Y) = D_{\mathrm{KL}}(p_{XY} \| p_X p_Y)$$

contracts under local stochastic channels.

It measures established correlation and does not derive an independent legacy-to-strict completion law.

10.6 Result

For each resource, 256 finite tests find no monotonicity violation. The theorems follow from exact data-processing principles.

[Proven] Monotonicity in the declared operational models.

[Conditional] Identification with FIN's abstract O108 grades.

10.7 New missing object

The newly isolated object is not another monotone. It is a typed family of encoding maps

$$\mathcal{E}_i : \text{FIN resource coordinate}_i \longrightarrow \text{operational state/channel pair}_i.$$

Without $\mathcal{E}_i$, an abstract resource count and a physical monotone remain different objects.

Chapter 11

P374 — an outer extension that does not derive phase

11.1 Construction

Define

$$F_\rho(z) = \sum_{d=0}^{\infty} K_{\text{strict}}(d)\rho^d z^d.$$

For $d \geq 1$,

$$|K_{\text{strict}}(d)| \leq \frac{1}{1+d^{9/5}} \leq \frac{1}{2}.$$

Therefore, on $|z| \leq 1$,

$$\left| \sum_{d\geq 1} K_{\text{strict}}(d)\rho^d z^d \right| \leq \frac{\rho}{2(1-\rho)}.$$

The constant term satisfies

$$K_{\text{strict}}(0) = \cos(13/80) \geq 1 - \frac{1}{2}(13/80)^2 = \frac{12631}{12800}.$$

11.2 Rouché threshold

Constant-term domination holds if

$$\frac{\rho}{2(1-\rho)} < \frac{12631}{12800}.$$

Equivalently,

$$0 < \rho < \frac{12631}{19031} \approx 0.6637065839945352.$$

Rouché's theorem gives zero-freeness on the closed disk. Absolute summability and a nonvanishing continuous boundary imply that F_ρ is outer, up to constant phase.

11.3 Why this is not the missing phase theorem

The construction inserts ρ. No strict artifact selects a unique ρ. More importantly, the coefficients of F_ρ already contain

$$\cos((743/4000)d + 13/80).$$

Minimum-phase reconstruction recovers phase already encoded in the coefficients. It does not derive ω or ϕ from the attenuation envelope.

Thus P374 proves:

- **[Proven]** an outer FIN analytic embedding exists;
- **[Refuted]** its existence alone supplies the frozen phase;
- **[Open]** a canonical FIN-derived analytic coordinate.

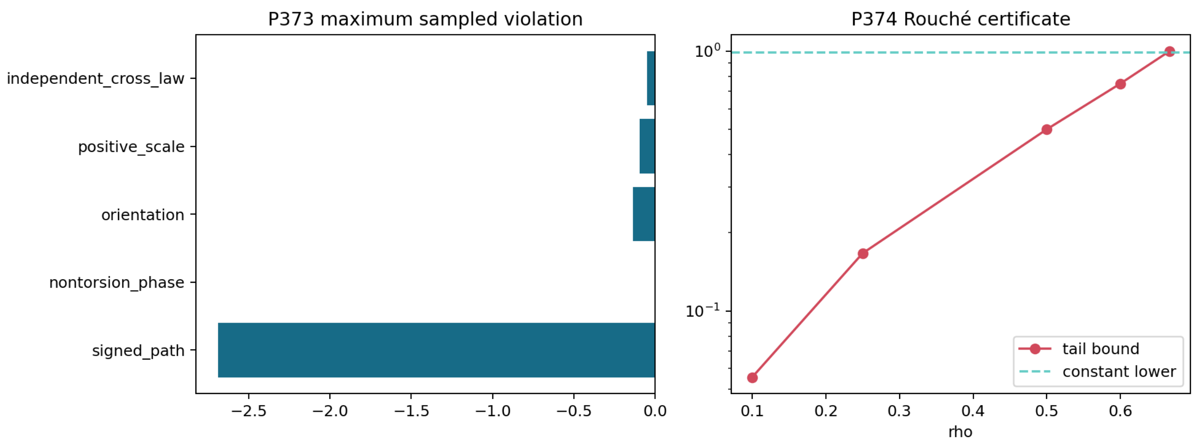

Figure 11.1: Operational data-processing semantics and the noncanonical zero-free outer analytic extension domain.

Chapter 12

P375–P376 — conditional sector and standards gates

12.1 Conditional electroweak package

No independently frozen manifest contains the conditional P348 prediction, sector axioms, external data, distinct roles, and one-shot unblinding.

P375: [Blocked by external evidence]

No legacy physical role is transferred.

12.2 Dimensional standards

No manifest supplies traceable

$$(\ell_*, \tau_*, \hbar_*)$$

with a covariance hash and independent roles.

P376: [Blocked by external evidence]

The P362 covariance pushforward is ready to consume such a record but cannot generate it.

Chapter 13

P377 — the exact universality boundary of FORC

13.1 Free resource syntax

The FORC resource object is

$$\mathbb{N}^5$$

under addition. Let M be any commutative monoid and choose five elements

$$m_1, \dots, m_5 \in M.$$

There is exactly one monoid homomorphism extending those generator images:

$$\Phi(r_1, \dots, r_5) = \sum_{i=1}^{5} r_i m_i.$$

Therefore $\mathbb{N}^5$ is the free commutative monoid on the five resource generators.

This proves the universal property of the A2/A4 bookkeeping skeleton.

13.2 Operational nonuniqueness

Consider the same zero resource grade:

$$r = (0, 0, 0, 0, 0).$$

Two operational models may assign:

$$p_A(1) = 0.25, \qquad p_B(1) = 0.75.$$

Their total-variation distance is

$$\mathrm{TV}(p_A, p_B) = 0.5.$$

The resource syntax does not distinguish them.

13.3 Result

[Proven] FORC is free and universal as an abstract commutative resource syntax.

[Refuted] FORC grades uniquely determine operational probabilities or a physical realization.

A5 is not merely an interpretation of A1–A4. It contains genuinely new empirical structure.

Chapter 14

P378 — reservoir process-tomography gate

No physical reservoir bundle contains:

- calibrated preparation maps;
- environmental controls;
- raw event records;
- a process-tomography reconstruction;
- a semigroup-versus-memory model comparison;
- independent custody and frozen hashes.

[**Blocked by external evidence**]

Synthetic evaluation of e^{-tA} is not reservoir engineering.

Chapter 15

Newly constructed theoretical objects

Object	**Definition**	**Resolves**	**Does not resolve**
O113 Proof-Carrying Moment Interface	Independent exact checker plus archived rational identities	Self-replay risk	Lean/Coq numerical reflection
O114 Seven-Atom Oscillatory Cell	Exact 12-variable Krawczyk enclosure	Improved primal upper bound	Physical meaning of signed mass
O115 Strict k-Distance Quotient Category	Maps preserving d modulo $k(d)$ equality	Maximal radial naturality	Legacy-to-strict completion
O116 Component-Gauge Identifiability Spectrum	Singular spectrum of transfer observation maps	Complex versus intensity calibration	Global hardware identifiability
O117 Adaptive-Angle Budget Certificate	Matching adaptive upper and parallel lower bounds	Ideal full-comb optimum to first threshold	Lossy/uncertain-clock comb
O118 Operational Resource-Semantics Pentagon	Five state/channel/ monotone triples	Candidate meanings of O108 coordinates	FIN encoding functors
O119 Damped Outer Generating Family	$F_\rho(z) = \sum K(d)\rho^d z^d$	Existence of an outer extension	Canonical ρ or phase source
O120 Free-Resource/ Empirical-Realization Split	Free $\mathbb{N}^5$ syntax plus probability countermodel	FORC universality boundary	Unique physical theory

Chapter 16

Falsification ledger

16.1 Surviving results

1. [Proven] A second exact implementation accepts the P351/P352 certificates.
2. [Proven] The P366 oscillatory enclosure is narrower on both sides.
3. [Proven] Radial naturality is exactly naturality on the k-distance quotient.
4. [Proven] On diameter at most four, the strict quotient reduces to ordinary distance.
5. [Moderate evidence] Complex transfer tomography is locally identifiable in the component model.
6. [Proven] The ideal unrestricted adaptive optimum equals the parallel optimum through the first perfect threshold.
7. [Proven] Five operational data-processing monotones exist.
8. [Proven] A noncanonical outer analytic FIN family exists.
9. [Proven] FORC's resource skeleton has a free universal property.
10. [Proven] Resource grades do not determine operational records.

16.2 Refuted promotions

1. Independent Python checking is not proof-assistant kernel checking.
2. A narrower signed cost is not a physical negativity theorem.
3. Kernel naturality is not arbitrary graph-homomorphism naturality.
4. Intensity-only calibration is not equivalent to complex transfer tomography.
5. P371 is not a lossy or clock-robust adaptive theorem.
6. Operational resource monotones are not automatically FIN encodings.
7. An outer embedding does not derive its own coefficients.
8. Free resource syntax is not unique empirical physics.
9. A blocked external gate is neither confirmation nor falsification.

Chapter 17

Present mathematical interpretation

The current FIN structure now separates into four precise levels:

$$\begin{array}{c}\text{spectral operator data}\\ \downarrow\\ \text{exact signed-resource and naturality theorems}\\ \downarrow\\ \text{conditional operational encodings}\\ \downarrow\\ \text{external calibrated realization and records.}\end{array}$$

The first two levels contain substantial mathematics. P371 also supplies an exact operational theorem once a particular ideal unitary comparison is declared.

The third level is not unique. P373 provides plausible data-processing semantics, while P377 proves that the abstract grades do not determine them.

The fourth level remains absent. This is the precise reason FIN is not yet a physical theory.

Chapter 18

Recommended Programs P379–P393

Program	Question	Deliverable	Probability of useful result
P379	Can the large rational certificates be reflected through Lean/Coq?	Small proof-certificate format and kernel-checked arithmetic	0.85
P380	Can exact contact/alternation reduce the oscillatory gap below 10^{-7}?	Certified primal-dual contact theorem	0.75
P381	Does the strict radial law have integer-distance collisions?	Global collision classification or explicit first collision	0.80
P382	Can one enriched naturality square relate legacy and strict kernels?	One typed completion atom or a categorical no-go	0.55
P383	Is the 132-parameter mesh globally identifiable modulo gauges?	Global gauge quotient and multitime calibration theorem	0.70
P384	Can a calibrated twelve-mode pilot be built?	Raw-event photonic transfer bundle	0.40
P385	Does the adaptive-angle theorem survive loss and dephasing?	Lossy comb upper/lower bounds	0.70
P386	How does bounded clock uncertainty deform the comb threshold?	Robust discrimination interval with nuisance clock	0.75
P387	Can one explicit FIN resource encoding functor be constructed?	Typed state/channel realization for one O108 coordinate	0.65
P388	Are any conversions between the five operational resources possible?	Conversion matrix, catalysts, or no-go theorems	0.60
P389	Can semigroup or RG composition select a canonical ρ?	Canonicality theorem or scale-orbit no-go	0.65
P390	Does QW survive an independent hold-out?	Custody-separated one-shot external test	0.50
P391	Can clock and dimensional standards be jointly calibrated?	Traceable $(\ell_*, \tau_*, \hbar_*)$ covariance package	0.45
P392	Can a physical reservoir realize the strict heat channel?	Process tomography with Markovian alternatives	0.35
P393	Can the conditional electroweak vector be tested blindly?	Frozen external sector test	0.20

18.1 P379 — arithmetic reflection

Design a compact certificate containing:

- fifth-root integer brackets;
- Taylor remainder parameters;
- Bernstein trees;
- rational inverse residuals;
- Krawczyk boxes;
- endpoint signs.

Lean or Coq should verify it without rerunning discovery.

18.2 P380 — exact contact geometry

The P366 gap remains approximately

$$7.50 \times 10^{-5}.$$

Recover the exact primal and dual contact set using alternation, moment semidefinite methods, or an interval KKT system.

18.3 P381 — global distance collisions

Classify integer pairs $d \neq e$ satisfying

$$K_{\mathrm{strict}}(d) = K_{\mathrm{strict}}(e).$$

Because the law oscillates while damping, global injectivity is not expected without proof. The first collision or a certified finite collision-free range directly controls the naturality category.

18.4 P382 — one enriched bridge square

Do not restart the generic legacy-to-strict audit. Select one typed carrier, one distance quotient, and one completion atom. Test a single commutative square:

$$\begin{array}{ccc} K_{\mathrm{legacy}}(G) & \longrightarrow & K_{\mathrm{strict}}(G) \\ \downarrow & & \downarrow \\ K_{\mathrm{legacy}}(H) & \longrightarrow & K_{\mathrm{strict}}(H). \end{array}$$

18.5 P383 — global photonic gauges

Local Jacobian rank does not exclude distant parameter aliases. Determine the exact gauge group of component losses/phases and design the smallest multitime complex calibration making the quotient identifiable.

18.6 P384 — physical pilot

This is an external program. Require:

- complex transfer calibration;
- raw timestamps;
- detector calibration;
- frozen protocol;
- distinct provider, registrar, and analyst;
- one-shot reporting.

18.7 P385 — lossy adaptive comb

Replace unitary calls by declared attenuation/dephasing channels. Seek:

- a contractive channel-angle upper bound;
- a feasible parallel or adaptive lower construction;
- a quantified gap;
- a dual certificate.

18.8 P386 — robust clock-comb theorem

For

$$\tau(t) \in \{\tau : \tau' > 0,\ |\tau''| \leq M\},$$

propagate the clock equivalence tube through

$$\sin\left(\frac{n\Delta\tau(t)}{2}\right).$$

Determine a guaranteed discrimination interval rather than a single uncalibrated threshold.

18.9 P387 — one realization functor

Choose exactly one resource coordinate and construct:

- FIN source object;
- operational state space;
- preparation;
- free channels;
- instrument;
- record;
- monotone equality or inequality.

This is the most direct internal attempt to instantiate A5 without claiming all physics.

18.10 P388 — conversions and catalysis

The five P373 resources have different codomains. Define typed conversion channels and determine whether any resource can generate another. A negative conversion matrix would sharpen the independence theorem; a positive entry would be a genuinely new bridge.

18.11 P389 — canonical outer scale

Test whether composition,

$$F_{\rho_1} \star F_{\rho_2},$$

semigroup laws, coarse graining, or an RG fixed point select one ρ. If every positive ρ remains on a scale orbit, prove the no-go.

18.12 P390–P393 — external falsification

These programs require independent organizations and physical records. Code may validate supplied manifests but may not create their evidential status.

18.13 Recommended immediate order

$$\boxed{\text{P379} \longrightarrow \text{P380} \longrightarrow \text{P381} \longrightarrow \text{P385} \longrightarrow \text{P387}}.$$

Chapter 19

Final conclusion

Programs P365–P378 sharpen the theory at three distinct boundaries.

First, the signed-moment results become more trustworthy and narrower. Independent exact implementation reproduces the certificates, and a seven-atom Krawczyk cell improves the full oscillatory enclosure.

Second, two previously open mathematical questions receive exact answers. Radial naturality lives on the k-distance quotient category. Ideal adaptive unitary discrimination is completely determined through the first perfect threshold by the relative-generator spectral diameter.

Third, the distinction between resource syntax and physics becomes a theorem. Five operational monotones can be constructed, but their connection to FIN requires separate encoding maps. FORC is universal as a free $\mathbb{N}^5$ resource syntax and demonstrably nonunique as an empirical realization.

The deepest surviving interpretation is therefore:

> *FIN is a dimensionless spectral-operator framework with exact signed resource obstructions, a well-defined carrier-naturality category, and conditional operational models. Its abstract resource syntax does not determine a unique physical realization.*

The smallest missing physical object remains an independently calibrated realization functor

$$\mathcal{R} : \mathsf{FINMath} \longrightarrow \mathsf{OperationalModel},$$

where

$$\begin{aligned}\mathsf{OperationalModel} := \{&\text{preparations, clocks, channels, environments,}\\ &\text{instruments, apparatus, probabilities, and records}\}.\end{aligned}$$

No spectral, categorical, or resource-universality theorem can manufacture the empirical content of $\mathcal{R}$. That content must be supplied and tested.

Chapter 20

Reproducibility

Run:

- MPLCONFIGDIR=/tmp/mpl-fin-1032 python3 fin_programs_365_378.py
- MPLCONFIGDIR=/tmp/mpl-fin-1032 python3 -m unittest -v test_fin_programs_365_378.py

Expected result: 14 tests passed, zero failed. The dependency-free Lean 4.28 structural core must compile. The P365 independent checker is executed by the main program.

Chapter 21

Suggested citation

Żuchowski, K. (2026). *FIN Research Programs P365–P378: Exact Certificate Diversity, Adaptive Discrimination, Operational Resource Semantics, and the Universality Boundary* (Release 10.32; Version 1.0.0) [Preprint]. Zenodo.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin-1032 \
python3 fin_programs_365_378.py

MPLCONFIGDIR=/tmp/mpl-fin-1032 \
python3 -m unittest -v test_fin_programs_365_378.py
```

Expected result: 14 tests passed, 0 failed. The dependency-free Lean 4.28 structural core must compile. P367, P370, P372, P375, P376, and P378 remain external admission gates; no synthetic record is accepted as laboratory, standards, hold-out, or custody evidence.

Suggested citation:

Żuchowski, K. (2026). *FIN Research Programs P365–P378: Exact Certificate Diversity, Adaptive Discrimination, Operational Resource Semantics, and the Universality Boundary* (Release 10.32; Version 1.0.0) [Preprint]. Zenodo.